# RL Objectives in DeepSeek-V3.2, GLM-5, MiMo-V2-Flash, and Kimi K2.5

One-row-per-model equation table. Equations are transcribed as faithfully as possible from the cited papers, with only line breaking and minimal notation grouping adjusted for layout.

**Shared notation.** $q, x$: prompt/input; $o_i, y_i$: sampled response $i$; $t$: token index; $|o_i|, |y_i|$: response length; $G, K$: number of sampled responses per prompt; $\pi_{\text{old}}$: rollout / old policy; $\pi_\theta$: current trainable policy; $\pi_{\text{ref}}$: frozen reference model; $R_i$ or $r(x, y_i)$: sequence reward; $\hat{A}_{i,t}$: advantage term; clip$(\cdot)$, Clip$(\cdot)$, and pop$(\cdot)$ denote the paper-specific clipping or gating operators.

| Model | Published RL / post-training objective (self-contained) |
|---|---|
| DeepSeek-V3.2 | $\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q\sim P(Q),\, \{o_i\}_{i=1}^{G}\sim\pi_{\text{old}}(\cdot\mid q)} \left[ \frac{1}{G}\sum_{i=1}^{G} \frac{1}{\lvert o_i\rvert} \sum_{t=1}^{\lvert o_i\rvert} \left( \min\left(r_{i,t}(\theta)\hat{A}_{i,t},\ \text{clip}(r_{i,t}(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_{i,t}\right) M_{i,t} - \beta D_{\text{KL}}\left(\pi_\theta(o_{i,t}) \,\|\, \pi_{\text{ref}}(o_{i,t})\right)\right)\right],$ <br> $r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\text{old}}(o_{i,t} \mid q, o_{i,<t})}, \qquad \hat{A}_{i,t} = R_i - \text{mean}(\mathbf{R}), \quad \mathbf{R} = \{R_1, \ldots, R_G\},$ <br> $M_{i,t} = \begin{cases} 0, & \hat{A}_{i,t} < 0 \text{ and } \frac{1}{\lvert o_i\rvert}\sum_{t=1}^{\lvert o_i\rvert} \log \frac{\pi_{\text{old}}(o_{i,t} \mid q, o_{i,<t})}{\pi_\theta(o_{i,t} \mid q, o_{i,<t})} > \delta, \\ 1, & \text{otherwise}, \end{cases}$ <br> $D_{\text{KL}}\left(\pi_\theta(o_{i,t}) \,\|\, \pi_{\text{ref}}(o_{i,t})\right) = \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\text{old}}(o_{i,t} \mid q, o_{i,<t})} \left( \frac{\pi_{\text{ref}}(o_{i,t} \mid q, o_{i,<t})}{\pi_\theta(o_{i,t} \mid q, o_{i,<t})} - \log \frac{\pi_{\text{ref}}(o_{i,t} \mid q, o_{i,<t})}{\pi_\theta(o_{i,t} \mid q, o_{i,<t})} - 1 \right).$ |
| GLM-5 | $\mathcal{L}(\theta) = -\mathbb{E}_{x\sim\mathcal{D},\, \{y_i\}_{i=1}^{G}\sim\pi^{\text{infer}}_{\theta_{\text{old}}}(\cdot\mid x)} \left[ \frac{1}{G}\sum_{i=1}^{G} \frac{1}{\lvert y_i\rvert} \sum_{t=1}^{\lvert y_i\rvert} \text{pop}(\rho_{i,t}, 1/\beta, \beta) \cdot \min\left(r_{i,t}\hat{A}_{i,t},\ \text{clip}(r_{i,t}, 1-\epsilon_{\text{low}}, 1+\epsilon_{\text{high}})\hat{A}_{i,t}\right)\right],$ <br> $\rho_{i,t} = \frac{\pi^{\text{train}}_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})}{\pi^{\text{infer}}_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})}, \qquad r_{i,t} = \frac{\pi^{\text{train}}_{\theta}(y_{i,t} \mid x, y_{i,<t})}{\pi^{\text{train}}_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})},$ <br> $\hat{A}_{i,t} = \frac{R_i - \text{mean}(R_1, \ldots, R_G)}{\text{std}(R_1, \ldots, R_G)}, \qquad \text{pop}(\rho_{i,t}, 1/\beta, \beta) = \begin{cases} \rho_{i,t}, & 1/\beta \le \rho_{i,t} \le \beta, \\ 0, & \text{otherwise}. \end{cases}$ |
| MiMo-V2-Flash | $\mathcal{L}_{\text{MOPD}}(\theta) = -\mathbb{E}_{x\sim\mathcal{D},\, y\sim\mu_\theta(\cdot\mid x)} \left[ \frac{1}{\lvert y\rvert} \sum_{t=1}^{\lvert y\rvert} w_t\, \hat{A}_{\text{MOPD},t}\, \log \pi_\theta(y_t \mid x, y_{<t}) \right],$ <br> $w_t(\theta) = \begin{cases} \text{sg}\left[\frac{\pi_\theta(y_t \mid x, y_{<t})}{\mu_\theta(y_t \mid x, y_{<t})}\right], & \epsilon_{\text{low}} \le \frac{\pi_\theta(y_t \mid x, y_{<t})}{\mu_\theta(y_t \mid x, y_{<t})} \le \epsilon_{\text{high}}, \\ 0, & \text{otherwise}, \end{cases}$ <br> $\hat{A}_{\text{MOPD},t} = \text{sg}\left[\log \frac{\pi^{\text{domain}}_x(y_t \mid x, y_{<t})}{\pi_\theta(y_t \mid x, y_{<t})}\right] + \alpha \hat{A}^{\text{ORM}}.$ |
| Kimi K2.5 | $L_{\text{RL}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}} \left[ \frac{1}{N}\sum_{j=1}^{K}\sum_{i=1}^{\lvert y_j\rvert} \left( \text{Clip}\left(\frac{\pi_\theta(y^i_j \mid x, y^{0:i}_j)}{\pi_{\text{old}}(y^i_j \mid x, y^{0:i}_j)},\ \alpha, \beta\right) \left(r(x, y_j) - \bar{r}(x)\right) - \tau \left(\log \frac{\pi_\theta(y^i_j \mid x, y^{0:i}_j)}{\pi_{\text{old}}(y^i_j \mid x, y^{0:i}_j)}\right)^2 \right)\right],$ <br> $\bar{r}(x) = \frac{1}{K}\sum_{j=1}^{K} r(x, y_j), \qquad N = \sum_{j=1}^{K} \lvert y_j\rvert.$ |

Source papers: DeepSeek-V3.2 (arXiv:2512.02556), GLM-5 (arXiv:2602.15763), MiMo-V2-Flash (arXiv:2601.02780), and Kimi K2.5 (arXiv:2602.02276).